

# CSCI 164: Machine Learning Report

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I analyzed the **Turkish Crowdfunding Startups** dataset (UCI ID 1025), which comprises 1,628 campaigns and **38** features spanning categorical platform details, project metrics, monetary targets, and free-text descriptions. After imputing a single missing region value, dropping raw date fields, and engineering three new features (log-transformed target and raised amounts plus numeric percent funded), we constructed a unified scikit-learn **ColumnTransformer** to one-hot encode categorical variables and standard-scale numeric ones.

Using this pipeline, we trained and tuned three classifiers via 5-fold **GridSearchCV** (F1 scoring):

- **Logistic Regression** ( $C=10$ )
- **k-Nearest Neighbors** ( $k=9$ , uniform weights)
- **Histogram-based Gradient Boosting** (learning\_rate=0.1, max\_depth=3, L2 regularization=0)

On a 30 % held-out test set, **logistic regression** emerged as the top performer with **F1 = 0.991** and **ROC-AUC = 0.997**, misclassifying only 3 out of 489 samples. **HistGradientBoosting** was closely followed (F1 = 0.987, AUC = 0.995), while **k-NN** trailed (F1 = 0.878, AUC = 0.970), likely due to the high-dimensional one-hot feature space.

Our results represent a substantial improvement over prior work by Kilinc & Aydin (2023), who reported **82.6 %** accuracy using **k-NN** on an earlier snapshot of the same data. We attribute this gain to targeted feature engineering (e.g., log-transforms), a dense encoding strategy, and rigorous hyperparameter tuning. These findings demonstrate that when paired with thoughtful preprocessing, simple, interpretable models can deliver near-perfect predictive power, offering crowdfunding platforms and entrepreneurs a reliable tool for campaign risk assessment and strategic planning.

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The other dataset we analyzed was the **Car Evaluation** dataset from the UCI Machine Learning Repository, which contains **1,728 instances** with **6 categorical features** describing cars (such as buying price, maintenance cost, number of doors, passenger capacity, luggage size, and safety rating) and **1 target variable** representing car acceptability (**unacc**, **acc**, **good**, or **vgood**).

All categorical features were **label-encoded** for model compatibility. Using this processed dataset, we trained and evaluated three machine learning models:

- **Logistic Regression** (baseline multiclass classifier)
- **k-Nearest Neighbors** (k=5 default and **k=7 after tuning**, uniform weights)
- **Linear Regression** (mapping class labels to ordinal values: unacc=0, acc=1, good=2, vgood=3)
- We also experimented with **KMeans clustering** to explore if natural groupings aligned with true labels.

We used GridSearchCV to tune the k-NN model.

The parameters we tested were:

- n\_neighbors: 3, 5, 7, 9
- weights: 'uniform' and 'distance'

The best combination found was **k=7** and **uniform weights**, achieving the highest cross-validated score (~90.88%). **Results**

On a 20% held-out test set, **Logistic Regression** achieved an accuracy of about **91%**, **k-NN** achieved a higher accuracy of about **93.6%**, making it the best performing model, **Linear Regression** had an **R<sup>2</sup> score of ~0.13**, indicating that regression wasn't ideal for this categorical prediction, **KMeans clustering** showed rough alignment with true classes but with no formal accuracy (unsupervised learning). Confusion matrices and classification reports showed that tuning **k-NN** improved precision and recall compared to the baseline models.

Prior studies, such as Jalali–Heravi et al. (2009), reported around **93% accuracy** using SVMs and ANNs on the same dataset. Our tuned **k-NN model** matched and slightly surpassed this benchmark (~93.6%), despite using simpler, more interpretable models. This shows that with careful preprocessing and hyperparameter tuning, even basic models like k-NN can achieve performance comparable to more complex methods on clean categorical datasets.